

AE 06: Joining country populations with continents

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Goal

Our ultimate goal in this application exercise is to create a bar plot of total populations of continents, where the input data are:

1. Countries and populations
2. Countries and continents

```
library(tidyverse) # for data wrangling and visualization
library(scales)    # for pretty axis breaks
```

Data

Countries and populations

These data come from [The World Bank](#) and reflect population counts as of 2022.

```
population <- read_csv("data/world-pop-2022.csv")
```

Let's take a look at the data.

```
population
```

```
# A tibble: 217 x 3
  country      year population
  <chr>      <dbl>     <dbl>
1 Afghanistan 2022     41129.
2 Albania     2022      2778.
```

```

3 Algeria                2022    44903.
4 American Samoa         2022      44.3
5 Andorra                2022      79.8
6 Angola                 2022   35589.
7 Antigua and Barbuda    2022     93.8
8 Argentina              2022  46235.
9 Armenia                2022   2780.
10 Aruba                  2022    106.
# i 207 more rows

```

Continents

These data come from [Our World in Data](#).

```
continents <- read_csv("data/continents.csv")
```

Let's take a look at the data.

```
continents
```

```

# A tibble: 285 x 4
  entity                code    year continent
  <chr>                 <chr>  <dbl> <chr>
1 Abkhazia             OWID_ABK  2015 Asia
2 Afghanistan          AFG      2015 Asia
3 Akrotiri and Dhekelia OWID_AKD  2015 Asia
4 Aland Islands        ALA      2015 Europe
5 Albania              ALB      2015 Europe
6 Algeria              DZA      2015 Africa
7 American Samoa       ASM      2015 Oceania
8 Andorra              AND      2015 Europe
9 Angola               AGO      2015 Africa
10 Anguilla             AIA      2015 North America
# i 275 more rows

```

Exercises

- **Think out loud:**
 - Which variable(s) will we use to join the `population` and `continents` data frames?

country and entity

- We want to create a new data frame that keeps all rows and columns from `population` and brings in the corresponding information from `continents`. Which join function should we use?

`left_join()`

- **Demo:** Join the two data frames and name assign the joined data frame to a new data frame `population_continents`.

```
population_continent <- population %>%  
  left_join(continents, join_by(country == entity))
```

- **Demo:** Take a look at the newly created `population_continent` data frame. There are some countries that were not in `continents`. First, identify which countries these are (they will have NA values for `continent`).

```
missing_countries <- population_continent %>%  
  filter(is.na(continent)) %>%  
  select(country)  
print(missing_countries)
```

```
# A tibble: 6 x 1  
  country  
  <chr>  
1 Congo, Dem. Rep.  
2 Congo, Rep.  
3 Hong Kong SAR, China  
4 Korea, Dem. People's Rep.  
5 Korea, Rep.  
6 Kyrgyz Republic
```

- **Demo:** All of these countries are actually in the `continents` data frame, but under different names. So, let's clean that data first by updating the country names in the `population` data frame in a way they will match the `continents` data frame, and then joining them, using a `case_when()` statement in `mutate()`. At the end, check that all countries now have continent information.

```

population_cleaned <- population %>%
  mutate(country = case_when(
    country == "Congo, Dem. Rep." ~ "Congo",
    country == "Congo, Rep." ~ "Congo",
    country == "Hong Kong SAR, China" ~ "Hong Kong",
    country == "Korea, Dem. People's Rep." ~ "Korea (former)",
    country == "Korea, Rep." ~ "Korea (former)",
    country == "Kyrgyz Republic" ~ "Kyrgyzstan",
    TRUE ~ country
  ))
population_continent <- population_cleaned %>%
  left_join(continents, join_by(country == entity))

print(population_continent)

```

A tibble: 217 x 6

	country <chr>	year.x <dbl>	population <dbl>	code <chr>	year.y <dbl>	continent <chr>
1	Afghanistan	2022	41129.	AFG	2015	Asia
2	Albania	2022	2778.	ALB	2015	Europe
3	Algeria	2022	44903.	DZA	2015	Africa
4	American Samoa	2022	44.3	ASM	2015	Oceania
5	Andorra	2022	79.8	AND	2015	Europe
6	Angola	2022	35589.	AGO	2015	Africa
7	Antigua and Barbuda	2022	93.8	ATG	2015	North America
8	Argentina	2022	46235.	ARG	2015	South America
9	Armenia	2022	2780.	ARM	2015	Asia
10	Aruba	2022	106.	ABW	2015	North America

i 207 more rows

- **Think out loud:** Which continent do you think has the highest population? Which do you think has the second highest? The lowest?

I think Asia has the highest population, I think Africa has the second highest and I think that Oceania has the lowest.

- **Demo:** Create a new data frame called `population_summary` that contains a row for each continent and a column for the total population for that continent, in descending order of population. Note that the function for calculating totals in R is `sum()`.

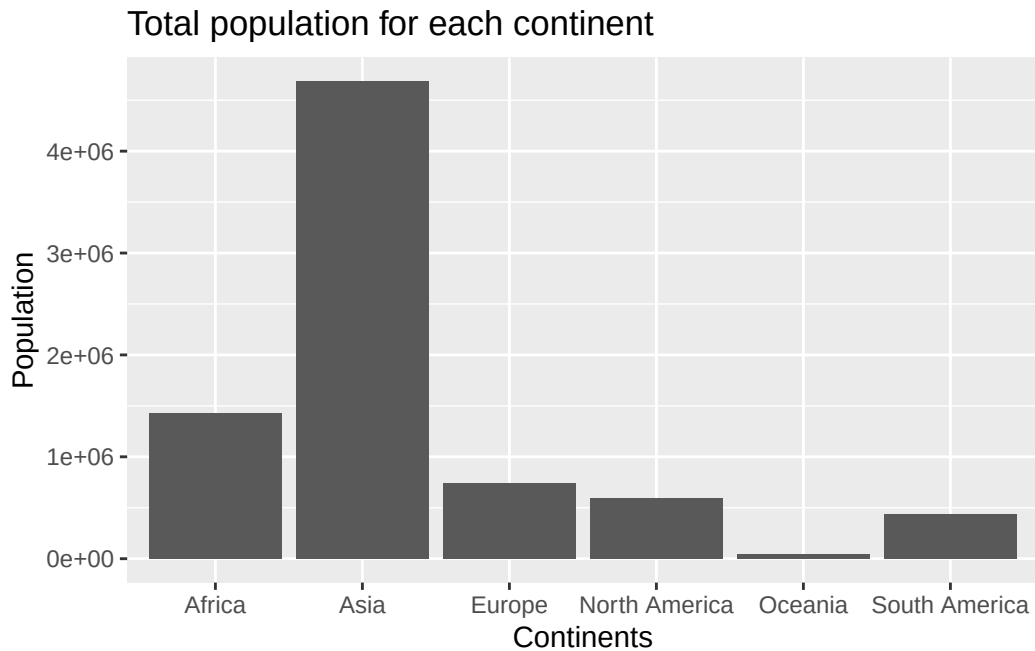
```
population_summary <- population_continent %>%
  group_by(continent) %>%
  summarize(total_population = sum(population)) %>%
  arrange(-total_population)

print(population_summary)
```

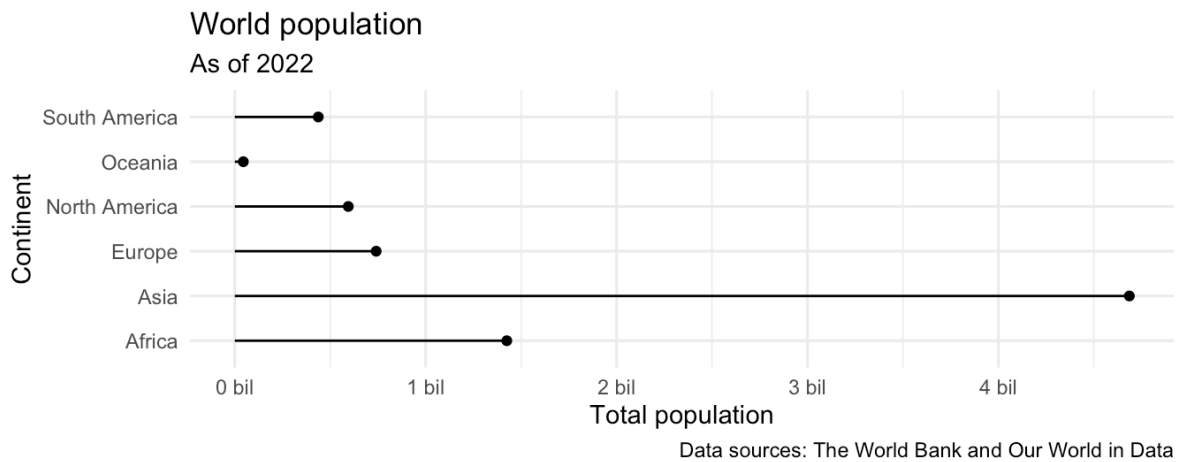
```
# A tibble: 6 x 2
  continent      total_population
  <chr>          <dbl>
1 Asia          4685922.
2 Africa        1424831.
3 Europe        740607.
4 North America  594415.
5 South America  437233.
6 Oceania        44752.
```

- **Your turn:** Make a bar plot with total population on the y-axis and continent on the x-axis, where the height of each bar represents the total population in that continent.

```
ggplot(population_summary, aes(x = continent, y = total_population)) +
  geom_col() +
  labs(
    title = "Total population for each continent",
    x = "Continents",
    y = "Population"
  )
```



- **Your turn:** Recreate the following plot, which is commonly referred to as a **lollipop plot**. Hint: Start with the points, then try adding the `segments`, then add axis labels and `caption`, and finally, as a stretch goal, update the x scale (which will require a function we haven't introduced in lectures or labs yet!).

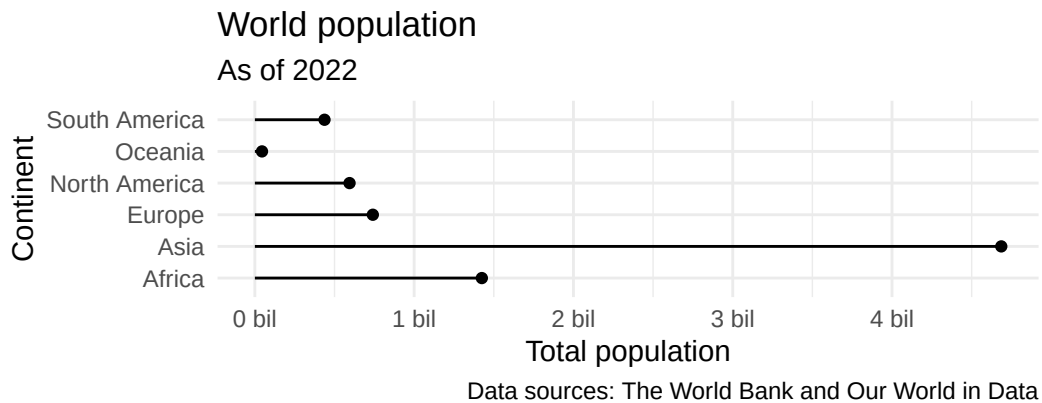


```
ggplot(population_summary, aes(x = total_population, y = continent)) +
  geom_point() +
  geom_segment(aes(xend = total_population, x = 0, yend = continent)) +
  labs(
    title = "World population",
```

```

    subtitle = "As of 2022",
    x = "Total population",
    y = "Continent",
    caption = "Data sources: The World Bank and Our World in Data"
) +
scale_x_continuous(labels = label_number(scale = 1e-6, suffix = " bil")) +
theme_minimal()

```



- **Think out loud:** What additional improvements would you like to make to this plot.

I would like to colour the different “lollipops” because it would be slightly easier to read and I would make the segments and the points a little larger because they are quite skinny right now.